Lecture 1 (September 1st)

Definition. (Tensor product) Let V and W be vector spaces over a common field. A tensor product of V and W is a pair

$$(V \otimes W, \otimes)$$

where $V \otimes W$ is a vector space over the aforementioned field and $\otimes : V \times W \to V \otimes W$ is a bilinear map (called the canonical tensor product map) such that for any linear map $f : V \times W \to Z$, there exists a unique linear map

$$\tilde{f} : V \otimes W \to Z$$

such that

$$f(v, w) = \tilde{f}(v \otimes w) = \tilde{f}(\otimes(v, w))$$

for all $v \in V$ and $w \in W$.

Definition. (Direct sum) Let V and W be vector spaces over a common field. The direct sum of V and W is defined as the vector space

$$V \oplus W := \{(v, w) \,|\, v \in V, w \in W\}$$

over the aforementioned common field.

Definition. (Fock space) Let $\mathcal{H}_1$ be a complex Hilbert space (A complete inner product space) that describes a single particle. The Fock space over $\mathcal{H}_1$, denoted $\mathcal{F}(\mathcal{H}_1)$, is defined as:

$$\mathcal{F}(\mathcal{H}_1) = \bigoplus_{n=0}^{\infty} (\mathcal{H}_1^{\otimes n})_{S,A} = \boldsymbol{C} \oplus \mathcal{H}_1 \oplus (\mathcal{H}_1 \otimes \mathcal{H}_1)_{S,A} \oplus \dots$$

Remark. There are some issues with combining special relativity (SR) and quantum mechanics (QM) naively:

(i) Special relativity treates t and $\mathbf{x}$ as equal compotents of a 4-vector while quantum mechanics quantises $\mathbf{x}$ but not t

(ii) Special relativity allows for creation and annhilation of particles whereas quantum mechanics doesn't accomodate such processes

The resultion is quantum field theory (QFT) which is done by quantising classical field theory.

Definition. (Special relativity) We deal with the Minkowski space denoted as $\boldsymbol{R}^{1,3}$, characterised by the metric

$$d(x, y) = \eta_{\mu\nu} x^\mu y^\nu$$

where $x^\mu = (ct, x, y, z)$ and $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$. We define automorphisms called the

Lorentz transformations (LT) as

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$$

where $\Lambda^{\mu}{}_{\rho} \eta_{\mu\nu} \Lambda^{\nu}{}_{\sigma} = \eta_{\rho\sigma}$. These transformations are those such that leave metric invariant, as seen below

$$ds'^2 = \eta_{\mu\nu} dx'^{\mu} dx'^{\nu} = \eta_{\mu\nu} \Lambda^{\mu}{}_{\rho} dx^{\rho} \Lambda^{\nu}{}_{\rho} dx^{\rho} = ds^2$$

In this course, we only deal with proper ($\det \Lambda = 1$) and orthochronous transformations $\Lambda^0{}_0 \geq 1$ (note that $|\Lambda^0{}_0| \geq 1$). In the language of matrices, we denote the (μ, ν) component of η^{-1} as $\eta^{\mu\nu}$ and of Λ^T as $(\Lambda^T)_{\mu}^{\nu}$. Using this, we can express the previous identity as

$$\eta = \Lambda^T \eta \Lambda$$

Meanwhile, Lorentz tensors (scalars, contravariant, and covariant) are defined as objects that transform accordingly. A general rank (p, g) tensor is then naturally defined as an object that transforms as

$$T'^{\mu'_1 \cdots \mu'_p}{}_{\nu'_1 \cdots \nu'_q}$$

Note that we can also define ∂_{μ} and ∂^{μ} that transform covariantly and contravariantly respectively. You'll need to check that η and its inverse lowers and raises indexes respectively. Depending on their norms, we can define vectors to be either time-like, space-like, and null/light-like.

Definition. (Representation of the Lorentz group) Are there representation of fields beyond tensors that transform a specific way under Lorentz transformations? Yes! Spinors! Consider the Lie algebra of the Lie group of Lorentz tranformations, where

$$\Lambda^{\mu}{}_{\nu} = \exp(i\omega^{\mu}{}_{\nu}) = \delta^{\mu}{}_{\nu} + i\omega^{\mu}{}_{\nu} + \mathcal{O}(\omega^2)$$

Using the definition of Lorentz transformations, we come to the conclusion that the algebra should satisfy

$$\omega_{\mu\nu} + \omega_{\nu\mu} = 0$$

There are 6 linearly independent 4×4 anti-symmetric matrices that satisfy such equation and we want to express them as a combination of transformation parameters and 6 independent Lorentz generators in vector representation.

$$\omega^{\mu}{}_{\nu} = \frac{1}{2} \alpha_{\rho\sigma} (M^{\rho\sigma})^{\mu}{}_{\nu}$$

where of course $M^{\rho\sigma} = -M^{\sigma\rho}$. With this freedom, we have

$$(M^{01})_{\mu\nu} = \begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

In general,

$$(M^{\rho\sigma})^{\mu}{}_{\nu} = -i(\eta^{\rho\mu}\delta^{\rho}{}_{\nu} - \eta^{\rho\mu}\delta^{\rho}{}_{\nu})$$

The commutation relations of Lorentz generators in vector representations thus become

$$[M^{\mu\nu}, M^{\rho\sigma}] = i(\eta^{\mu\rho}M^{\nu\sigma} + \eta^{\nu\rho}M^{\mu\rho} - \eta^{\mu\rho}M^{\nu\rho} - \eta^{\nu\rho}M^{\mu\rho})$$

Which is called the Lorentz algebra. The Lorentz algebra is more fundamental than the specific form of generators in the specific representation. General Lorentz transformations in temrs of Lorentz generators thus become

$$\Lambda = \exp\left(\frac{i}{2}\alpha_{\mu\nu}M^{\mu\nu}\right)$$

$\alpha_{\mu\nu} \in \boldsymbol{R}$ $M^{\mu\nu} = (M^{\mu\nu})^{\dagger}$ thus $\Lambda^{\dagger} = \Lambda$. A natural question is – what do $M^{\mu\nu}$ look like in the spinor representation?

Lecture 2 (September 3rd)

Recall. We have learned the Lorentz algebra that satisfies the commutator relations

$$[M^{\mu\nu}, M^{\rho\sigma}] = i\Big(\eta^{\mu\rho}M^{\nu\sigma} + \eta^{\nu\sigma}M^{\mu\rho} - \eta^{\mu\sigma}M^{\nu\rho} - \eta^{\nu\rho}M^{\mu\sigma}\Big)$$

where

$$\Lambda = \exp\left[\frac{i}{2}\alpha_{\mu\nu}M^{\mu\nu}\right]$$

Proposition. (Irreducible representation of the Lorentz algebra) We now want seek to find a specific representation of the Lorentz algebra, suitable for distinguishing how different particles are acted by them. First, we find the individual components of the matrix to be denoted by

$$J^i = \frac{1}{2}\epsilon^{ijk}M^{jk} \qquad K^i = M^{0i}$$

The Lie algebra becomes

$$[J^i, J^j] = i\epsilon^{ijk}J^k \qquad [J^i, K^j] = i\epsilon^{ijk}K^k \qquad [K^i, K^j] = -i\epsilon^{ijk}J^k$$

From here, we can complexify the Lorentz algebra and express

$$J^{\pm i} = \frac{1}{2}(J^i \pm iK^i)$$

What about the commutator relations?

$$[J^{\pm i}, J^{\pm j}] = i\epsilon^{ijk}J^k \qquad [J^{\pm i}, J^{\mp j}] = 0$$

Note that through this complexification process, we are essentially expressing the Lorentz algebra, the tangent space of the Lorentz group at e, through the real $\mathfrak{su}(2)$ component and an imaginary $\mathfrak{su}(2)$ component given the homomorphism

$$\mathfrak{so}(1,3) \simeq \mathfrak{su}(2) \oplus \mathfrak{su}(2)$$

With all the information above, we can simply write the general Lorentz transformation through the parameters

$$\vec{\alpha} = (\alpha_{23}, \alpha_{31}, \alpha_{12}) \quad \vec{\beta} = (\alpha_{01}, \alpha_{02}, \alpha_{03})$$

where

$$\Lambda = \exp\left[\frac{i}{2}\alpha_{\mu\nu}M^{\mu\nu}\right] = \exp\left[i\vec{\alpha}\cdot\vec{J} + i\vec{\beta}\cdot\vec{K}\right] = \exp\left[i(\vec{\alpha} - i\vec{\beta})\cdot\vec{J}^{+}\right]\exp\left[i(\vec{\alpha} + i\vec{\beta})\cdot\vec{J}^{-}\right]$$

Definition. ((j_+, j_-)-representation of the Lorentz transform) Note that by definition, a representation of a Lorentz algebra is a Lie algebra homomorphism

$$\phi : \mathfrak{so}(3,1)_{\boldsymbol{C}} \to \mathrm{End}(V)$$

where V is the representation space. Note that a Lie algebra homomorphism is a map between two Lie algebras that respects both the vector space structure and the bracket.

$$\phi([X,Y]) = [\phi(X), \phi(Y)]$$

We now create these target vector spaces using ladder operators. We now define a (j_+, j_-) representation space of the Lorentz transformations to be the vector space with the basis

$$\{m_+, m_- \mid m_\pm = -j_\pm, -j_\pm + 1, \ldots, j_\pm\}$$

(here, $j_\pm \in \boldsymbol{Z}_{\geq 0}/2$) constructed through the ladder operators

$$J^+_\pm = J^{+1} \pm iJ^{+2} \qquad J^-_\pm = J^{-1} \pm iJ^{-2}$$

satisfying

$$J^+_\pm|m_+, m_-\rangle \sim |m_+ \pm 1, m_-\rangle \qquad J^-_\pm|m_+, m_-\rangle \sim |m_+, m_- \pm 1\rangle$$

We would also have

$$\sum_{i=1}^{3}(J^\pm)^2|m_+, m_-\rangle = j_\pm(j_\pm + 1)|m_+, m_-\rangle$$

Example. (1D (0,0)-representation as spin-0 scalars) We see that for this representation, all Lorentz transforms are of the form

$$\Lambda|_{(0,0)} = \mathbf{1}$$

and that the vector space elements $|0,0\rangle = \phi(x)$ transform as

$$\phi'(x') = \Lambda|_{(0,0)}\phi(x) = \phi(x)$$

Example. (2D (1/2,0)-representation as spin-1/2 left-chiral spinors) We see that for this representation, all Lorentz transforms are of the form

$$\Lambda|_{(1/2,0)} = \exp\left[i(\vec{\alpha} - i\vec{\beta} \cdot \vec{J}^{+})\right]$$

As we are dealing with a 2D complex vector space, an arbitrary transformation is a linear combination of the Pauli matrices, and in matrix notation,

$$l_{+} = \exp\left[i(\vec{\alpha} - i\vec{\beta}) \cdot \frac{\vec{\sigma}}{2}\right]$$

and vector space elements $| \pm j, 0\rangle = \psi_{+}(x)$ transform as

$$\psi'_{+}(x') = \Lambda|_{(1/2,0)}\psi_{+}(x) = l_{+}\psi_{+}(x)$$

Example. (2D (0,1/2)-representation as spin-1/2 right-chiral spinors) We see that for this representation, all Lorentz transforms are of the form

$$\Lambda|_{(0,1/2)} = \exp\left[i(\vec{\alpha} - i\vec{\beta} \cdot \vec{J}^{-})\right]$$

As we are dealing with a 2D complex vector space, an arbitrary transformation is a linear combination of the Pauli matrices, and in matrix notation,

$$l_{-} = \exp\left[i(\vec{\alpha} - i\vec{\beta}) \cdot \frac{\vec{\sigma}}{2}\right]$$

and vector space elements $|0, \pm J\rangle = \psi_{-}(x)$ transform as

$$\psi'_{-}(x') = \Lambda|_{(0,1/2)}\psi_{-}(x) = l_{-}\psi_{-}(x)$$

Definition. (Chirality and helicity) We call a vector left-chiral or right-chiral based on which $SU(2)$ factor of the Lorentz group it transforms by. Helicity is a property of fields.

Lemma. We find that

(i) Choosing how the left-chiral spinors transform automatically dictate how the right-chiral spinors transform, as

$$(\vec{J}^{\pm})^{\dagger} = \vec{J}^{\mp}$$

(ii) The map between left and right-chiral spinors manifest through the fact that

$$(l_\pm)^\dagger = (l_\mp)^{-1} \qquad \& \qquad (l_\pm)^* = \sigma_2 l_\mp \sigma_2$$

as $SU(2)$ elements. We see that

$$\sigma_2 \psi'_\pm(x')^* = l_\mp \sigma_2 \psi_\pm(x)^*$$

Therefore, conjugation is the exact map between left and right-chiral spinors.

Definition. (Invariant tensors) We define

$$\begin{cases} \sigma^\mu = (I_2, \vec{\sigma}) \\ \bar{\sigma}^\mu = (I_2, -\vec{\sigma}) \end{cases}$$

which leads

$$\begin{cases} (l_-)^+ \sigma^\mu l_- = \Lambda^\mu{}_\nu \sigma^\nu \\ (l_+)^+ \bar{\sigma}^\mu l_+ = \Lambda^\mu{}_\nu \bar{\sigma}^\nu \end{cases}$$

Therefore the Lorentz generators for the $\{(1/2,0),(0,1/2)\}$ representations in terms of σ^μ and $\bar{\sigma}^\mu$ can be expressed as

$$\begin{cases} (\frac{1}{2}, 0) : M^{0i} = -\frac{i}{2}\sigma^i, M^{ij} = \frac{1}{2}\varepsilon^{ijk}\sigma^k \end{cases}$$

Lecture 3 (September 8th)

Definition. (Grassmann numbers) Grassman numbers are numbers, contrarily to complex numbers, that describe fermions. Grassmann variables are tuplets

$$\{\theta_1, \dots, \theta_n\}$$

that satisfy

$$\{\theta_i, \theta_j\} = \theta_i\theta_j + \theta_j\theta_i = 0$$

The Grassmann algebra is defined as

$$\Lambda(v) = \boldsymbol{C} \oplus V \oplus V^2 \oplus \dots \oplus V^n$$

where V is an n-dimensional vector space with the basis $\{\theta_i\}$ over the field of complex numbers. Basic operations are defined as

(i) Addition $\theta_i + \theta_j$

(ii) Complex multiplication $c\theta_i$ $(c \in \boldsymbol{C})$

(iii) Product $\theta_i\theta_j$

The general element of the Grassmann algebra is called the Grassmann number,

$$\begin{aligned} f(\theta) =& f_0 + \sum_{i_1=1}^{n} f_{i_1}\theta_{i_1} + \frac{1}{2}\sum_{i_1,i_2=1}^{n} f_{i_1 i_2}\theta_{i_1}\theta_{i_2} + \dots \\ =& \sum_{k=0}^{n}\frac{1}{k!}\sum_{i_1,\dots,i_k=1}^{n} f_{i_1\dots i_k}\theta_{i_1}\dots\theta_{i_k} \end{aligned}$$

Definition. (Differentiation of Grassmann variables)

$$\frac{\partial\theta_j}{\partial\theta_i} = \delta_{ij} \qquad \frac{\partial\mathbf{1}}{\partial\theta_i} = 0 \qquad \frac{\partial}{\partial\theta_i}(\theta_j\theta_k) = \delta_{ij}\theta_k$$

Definition. (Integration of Grassmann variables)

$$\int d\theta\, f(\theta) = \frac{\partial}{\partial\theta} f(\theta)$$

Notice that once we change the integration measure from $\theta'_i = A_{ij}\theta_j$,

$$\begin{aligned} \int d\theta_1 \dots d\theta_n\, f(\theta) =& \frac{\partial}{\partial\theta_1}\cdots\frac{\partial}{\partial\theta_n} f(\theta) \\ =& A_{i_1 1}\dots A_{i_n n}\frac{\partial}{\partial\theta'_{i_1}}\cdots\frac{\partial}{\partial\theta'_{i_n}} f(A^{-1}\theta') \\ =& \varepsilon_{i_1\dots i_n} A_{i_1 1}\dots A_{i_n n}\frac{\partial}{\partial\theta'_1}\cdots\frac{\partial}{\partial\theta'_n} f(A^{-1}\theta') \\ =& \det(A)\int d\theta'_1\dots d\theta'_n\, f(A^{-1}\theta') \end{aligned}$$

which gives us

$$d\theta'_1\dots d\theta'_n = (\det A)^{-1} d\theta_1\dots d\theta_n$$

as

$$\frac{\partial}{\partial\theta_i} = \frac{\partial\theta'_j}{\partial\theta_i}\frac{\partial}{\partial\theta'_j}$$

Definition. (Gaussian integral wth respect to two sets of Grassman variables)

$$\begin{aligned}
&\int d\bar{\theta}_1 d\theta_1 \dots d\bar{\theta}_n d\theta_n \, e^{-\sum_{i,j=1}^n \bar{\theta}_i M_{ij} \theta_j} \\
&= \int d\bar{\theta}_1 d\theta'_1 \dots d\bar{\theta}_n d\theta'_n (\det M) e^{o \sum_{i=1}^n \bar{\theta}_i \theta'_i} \\
&= \det M \prod_{i=1}^n \int d\bar{\theta}_i d\theta'_i \, e^{-\bar{\theta}_i \theta'_i} \\
&= \det M \prod_{i=1}^n \int d\bar{\theta}_i d\theta'_i (1 - \bar{\theta}_i \theta'_i) = \det M
\end{aligned}$$

(where $M_{ij}\theta_j = \theta'_i$)

Definition. (Functional derivatives) Let X be a space of admissible functions $y : [a,b] \to \boldsymbol{R}$ like the Banach space $L^2([a,b])$ or $C^1([a,b])$. A functional is a mapping $F : X \to \boldsymbol{R}$ defined as

$$\frac{\delta F[f]}{\delta f(x_0)} = \lim_{\varepsilon \to 0} \frac{F[f(x) + \varepsilon \delta(x - x_0)] - F[f(x)]}{\varepsilon}$$

Example. (Examples of functional derivatives) We have

(i) $\int f^n$ gives

$$\frac{\delta F[f]}{\delta f(z)} = n(f(z))^{n-1}$$

(ii) $\int (f')^n$ gives

$$\frac{\delta F[f]}{\delta f(z)} = -nf^{(n)}(z)$$

(iii) $\int dx \, K(y,x) f(x)$ gives

$$\frac{\delta F[f]}{\delta f(z)} = K(y,z)$$

(iv) $f(y)$ gives

$$\frac{\delta F[f]}{\delta f(z)} = \delta(y - z)$$

Classical Field Theory

Theorem. (Lagrangian formalism) Classically, the action S of a system is written as

$$S[x, \dot{x}] = \int_{t_i}^{t_f} dt \, L(x_i, \dot{x}_i)$$

For a field, we write

$$S[\phi, \dot{\phi}] = \int_{t_i}^{t_f} dt\, L[\phi, \dot{\phi}]$$

where $\phi = \phi(t, \vec{x}) = \phi(x)$. The variation of this action is given as

$$\delta S[\phi, \dot{\phi}] = \int dt \int d^3\vec{x} \Big[\frac{\delta L}{\delta \phi(x)}\delta\phi(x) + \frac{\delta L}{\delta \dot{\phi}(x)}\delta\dot{\phi}(x)\Big]$$

Through the application of Hamilton's principle (the variation of the action is zero for an arbitrary variation of the field such that $\delta\phi(t_i, \vec{x}) = \delta\phi(t_f, \vec{x}) = 0$) we obtain

$$\begin{aligned} 0 &= \int d^4x \Big(\frac{\delta L}{\delta \phi(x)}\delta\phi(x) + \frac{\delta L}{\delta \dot{\phi}(x)}\frac{\partial}{\partial t}\delta\phi(x)\Big) \\ &= \int d^4x \Big(\frac{\delta L}{\delta \phi(x)} - \frac{\partial}{\partial t}\frac{\delta L}{\delta \dot{\phi}(x)}\Big)\delta\phi(x) + \int d^4x\, \frac{\partial}{\partial t}\Big(\frac{\delta L}{\delta \dot{\phi}(x)}\delta\phi(x)\Big) \end{aligned}$$

where the second boundary terms vanish to zero. The Lagrangian in "local" relativistic classical field theory leads to the concept of a Lagrangian density.

$$S[\phi, \dot{\phi}] = \int dt\, L[\phi, \dot{\phi}] = \int d^4x\, \mathcal{L}(\phi(x), \dot{\phi}(x), \nabla\phi(x)) = \int d^4x\, \mathcal{L}(\phi(x), \partial_\mu\phi(x))$$

Alternative expressions in local classical field theory include

$$\frac{\delta L}{\delta \phi} = \frac{\partial \mathcal{L}}{\partial \phi} - \nabla\Big(\frac{\partial \mathcal{L}}{\partial(\nabla\phi)}\Big) \qquad \frac{\delta L}{\delta \dot{\phi}} = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$$

and the Euler-Lagrangian equations as

$$0 = \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\Big)$$

as

$$\begin{aligned} 0 &= \int d^4x \Big[\frac{\partial \mathcal{L}}{\partial \phi}\, d\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi)\Big] \\ &= \int d^4x \Big(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\Big)\delta\phi + \int d^4x\, \partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi\Big)\Big) \end{aligned}$$

assuming that $\phi, \partial_\mu\phi \to 0$ fast enough as $|\vec{x}| \to \infty$.

Theorem. (Hamiltonian formulation) The canonically conjugate field is defined as

$$\pi(x) = \frac{\delta L}{\delta \dot{\phi}(x)} = \frac{\partial \mathcal{L}}{\partial \dot{\phi}(x)}$$

The Hamiltonian is then defined as

$$H[\phi, \pi] = \int d^3\vec{x}\, \Big(\pi(x)\dot{\phi}(x) - \mathcal{L}\Big) = \int d^3x\, \mathcal{H}$$

The Eular-Lagrange equations then become Hamilton's equations of motion

$$\dot{\phi}(x) = \frac{\delta H}{\delta \pi(x)} \qquad \dot{\pi}(x) = -\frac{\delta H}{\delta \phi(x)}$$

Lecture 4 (September 10th)

Definition. (Poisson bracket) The Poisson bracket for two functions $F = F[\phi, \pi]$ and $G = G[\phi, \pi]$ is defined as

$$\{F, G\}_{PG} = \int d^3\vec{x} \Big(\frac{\delta F}{\delta \phi(x)} \frac{\delta G}{\delta \pi(x)} - \frac{\delta F}{\delta \pi(x)} \frac{\delta G}{\delta \phi(x)} \Big)$$

Notice that using the Poisson bracket, Hamilton's equation of motion becomes

$$\begin{cases} \dfrac{d\phi}{dt} = \dfrac{\delta H}{\delta \pi} = \{\phi, H\}_{PB} \\[2ex] \dfrac{d\pi}{dt} = -\dfrac{\delta H}{\delta \phi} = \{\phi, H\}_{PB} \end{cases}$$

Then, for an arbitrary functional F,

$$\begin{aligned} \frac{dF}{dt} &= \int d^3\vec{x} \Big(\frac{\delta F}{\delta \phi(x)} \dot{\phi}(x) + \frac{\delta F}{\delta \pi(x)} \dot{\pi}(x) \Big) \\ &= \int d^3\vec{x} \Big(\frac{\delta F}{\delta \phi} \frac{\delta H}{\delta \pi} + \frac{\delta F}{\delta \pi} \Big(-\frac{\delta H}{\delta \phi} \Big) \Big) = \{F, G\}_{PB} \end{aligned}$$

We have some special cases,

$$\begin{aligned} \{\phi(t, \vec{x}), \phi(t, \vec{y})\}_{PB} =& 0 = \{\pi(t, \vec{x}), \pi(t, \vec{y})\}_{PB} \\ \{\phi(t, \vec{x}), \pi(t, \vec{y})\}_{PB} =& \int d^3 z \, \frac{\delta \phi(t, \vec{x})}{\delta \phi(t, \vec{z})} \frac{\delta \pi(t, \vec{y})}{\delta \pi(t, \vec{z})} = \int d^3 z \, \delta^{(3)}(\vec{x} - \vec{z}) \delta^{(3)}(\vec{y} - \vec{z}) = \delta^{(3)}(\vec{x} - \vec{y}) \end{aligned}$$

Definition. (Spin-0 Klein-Gordan Fields) The Klein-Gordan (KG) action is given as

$$S_{KG} = \int d^4x \, \underbrace{\Big(-\frac{1}{2} \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} m^2 \phi^2 \Big)}_{\mathcal{L}_{KG}}$$

The Klein-Gordan equation that comes from this action is

$$0 = \frac{\partial \mathcal{L}_{KG}}{\partial \phi} - \partial_\mu \Big(\frac{\partial \mathcal{L}_{KG}}{\partial(\partial_\mu \phi)} \Big) = -m^2 \phi - \partial_\mu(-\partial^\mu \phi) = (\partial^2 - m^2)\phi$$

here, $\partial^2 = \partial^\mu \partial_\mu = \Box$. What are the solutions to the Klein-Gordan equation? The simplest case is given as a plane wave solution

$$\phi_{\pm,\vec{p}}(x) = A_\pm e^{i(\vec{p}\cdot\vec{x}\pm E_p t)}$$

where $E_p = \sqrt{\vec{p}^2 + m^2}$. Is it genuinely a solution? Notice that

$$(\partial^2 - m^2)\phi_{\pm,\vec{p}}(x) = (-\partial_t^2 + \nabla^2 - m^2)\phi_{\pm,\vec{p}} = (E_p^2 - \vec{p}^2 - m^2)\phi_{\pm,\vec{p}} = 0$$

The general soltuion is then given by a Fourier expansion,

$$\phi(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}} \Big(A_-(\vec{p})e^{i(\vec{p}\cdot\vec{x}-E_p t)} + A_+(-\vec{p})e^{-i(\vec{p}\cdot\vec{x}-E_p t)} \Big)$$

where we will now write $\vec{p}\cdot\vec{x} - E_p t = p_\mu x^\mu$. Imposing the condition that $\phi(x) = \phi^*(x)$,

$$\phi(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}} \Big(a(\vec{p})e^{ip_\mu x^\mu} + a^*(\vec{p})e^{-p_\mu x^\mu} \Big)$$

The canonical conjugate field is then automatically given as

$$\begin{aligned}\pi(x) =& \frac{\partial \mathcal{L}_{KG}}{\partial \dot{\phi}(x)} = \dot{\phi}(x) \\ =& \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}} (-iE_p)(a^*(\vec{p})e^{ip\cdot x} - a(\vec{p})^* e^{-ip\cdot x}\end{aligned}$$

The Hamiltonian density becomes

$$\mathcal{H}_{KG} = \pi\dot{\phi} - \mathcal{L}_{KG} = \frac{1}{2}(\pi^2 + (\nabla\phi)^2 + m^2\phi^2)$$

The Poisson brackets become

$$\begin{cases} \{\phi(t,\vec{x}), \phi(t,\vec{y})\}_{PB} = \{\pi(t,\vec{x}), \pi(t,\vec{y})\}_{PB} = 0 \\ \{\phi(t,\vec{x}), \pi(t,\vec{y})\}_{PB} = \delta^{(3)}(\vec{x} - \vec{y}) \end{cases}$$

Definition. (Spin-1/2 Dirac fields) There is no first order, Lorentz covariant equation of motion for spin-0 fields, but we can find one for spin-1/2 fields. We begin from the Weyl equations of motion:

$$\begin{cases} i\bar{\sigma}^\mu \partial_\mu \psi_+ + m\sigma_2 \psi_+^* = 0 \\ i\sigma^\mu \partial_\mu \psi_- + m\sigma_2 \psi_-^* = 0 \end{cases}$$

Notice that without the absence of math, they require both left and right chirals. A problem with the equations is that there is no $U(1)$ symmetry.

$$\psi_\pm \to \psi'_\pm = e^{i\theta_\pm}\psi_\pm$$

Coupling the two Weyl equations of motion with doubled degrees of freedom, we have

$$\begin{cases} 0 = i\bar{\sigma}^\mu \partial_\mu \psi_+ + m\psi_- \\ 0 = i\sigma^\mu \partial_\mu \psi_- + m\psi_+ \end{cases} \iff \left(\begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \partial_\mu - m \right) \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = 0$$

The Dirac action that yields this equation is

$$S_D = \int d^4x \, \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi$$

where $\gamma^\mu \partial_\mu = \not{\partial}$. The dirac equation is therefore

$$0 = \frac{\partial \mathcal{L}_D}{\partial \bar{\psi}} - \partial_\mu \Big(\frac{\partial \mathcal{L}_D}{\partial(\partial_\mu \bar{\psi})} \Big) = (i\not{\partial} - m)\psi$$

A natural equation would be to ask what would happen for $\bar{\psi}$. The elementary solution as a plane wave is given as

$$\psi_{\pm,\vec{p}}(x) = u_\pm(\vec{p}) e^{\pm i p_\mu x^\mu}$$

where $u_\pm(\vec{p})$ is expected to be a 4×1 matrix with $(\not{p} \pm m)u_\pm(\vec{p}) = 0$. We can check that for our elementary solution.

$$(i\not{\partial} - m)\psi_{\pm,\vec{p}}(\vec{x}) = (i(\pm i\gamma^\mu p_\mu) - m)\psi_{\pm,\vec{p}} = \mp(\not{p} \pm m)\psi_{\pm,\vec{p}} = 0$$

Note that also, $p^\mu p_\mu + m^2 = 0$.

$$\begin{aligned} 0 = &- (\gamma^\nu p_\nu \mp m)(\gamma^\mu p_\nu \pm m)u_\pm(\vec{p}) = -(\gamma^\nu \gamma^\mu p_\nu p_\mu - m^2)u_\pm(\vec{p}) \\ = &- \Big(\frac{1}{2}\{\gamma^\nu, \gamma^\mu\} p_\nu p_\mu - m^2 \Big) u_\pm(\vec{p}) = -(-p^\mu p_\mu - m^2)u_\pm(\vec{p}) = 0 \end{aligned}$$

where the commutators equal $-2\eta^{\mu\nu}$. Using the Dirac basis we can express the four independent solutions for $u_\pm$ (2 for each sign) as

$$0 = (\not{p} \pm m)u_\pm(\vec{p}) \propto \begin{pmatrix} \pm m - E_p & \vec{p} \cdot \vec{\sigma} \\ -\vec{p} \cdot \vec{\sigma} & \pm m + E_p \end{pmatrix} \begin{pmatrix} \phi_\pm \\ \chi_\pm \end{pmatrix}$$

This implies that

$$\begin{cases} \chi_+ = & \dfrac{\vec{p} \cdot \vec{\sigma}}{m + E_p} \phi_+ \\ \phi_- = & \dfrac{\vec{p} \cdot \vec{\sigma}}{m + E_p} \chi_- \end{cases}$$

The two independent u_+ become (for $i = 1, 2$)

$$u_+^i(\vec{p}) = \sqrt{m+E_p}\begin{pmatrix} \phi_+^i \\ \dfrac{\vec{p}\cdot\vec{\sigma}}{m+E_p}\phi_+^i \end{pmatrix}$$

and two independent for u_- become

$$u_-^i(\vec{p}) = \sqrt{m+E_p}\begin{pmatrix} \dfrac{\vec{p}\cdot\vec{\sigma}}{m+E_p}\chi_-^i \\ \chi_-^i \end{pmatrix}$$

with

$$\begin{cases} (\phi_+^i)^\dagger \phi_+^j = \delta^{ij} \\ \sum_{i=1}^2 \phi_+^i(\phi_+^i)^\dagger = I_2 \end{cases}$$

and the same for χ_-^i. A good example of vectors that do this are

$$\phi_+^j = \{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\}$$

Remark. (Properties of $u_\pm^i(\vec{p})$) We have orthonormality

$$\bar{u}_\pm^i(\vec{p})u_\pm^j(\vec{p}) = \pm 2m\delta^{ij}$$

and

$$\bar{u}_\pm^i(\vec{p})u_\mp^j(\vec{p}) = 0$$

and the spin state sum

$$\sum_{i=1}^2 u_\pm^i \bar{u}_\pm^i(\vec{p}) = -(\not{p} \mp m)$$

Other useful properties include

$$\begin{cases} u_\pm^i{}^\dagger(\vec{p})u_\pm^j(\vec{p}) = 2E_p\delta^{ij} \\ u^i) \pm (\vec{p})^\dagger u_\mp^j(-\vec{p}) = 0 \end{cases}$$

The general solution then

Lecture 5 (September 15th)

Definition. (Spin-1 Maxwell fields) The Maxwell action is given as

$$S_M = \int d^4x \, \underbrace{\Big(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}\Big)}_{\mathcal{L}_M}$$

Note that for a vector field $A_\mu(x)$, the field strength is defined as

$$F_{\mu\nu} = 2\partial_{[\mu}A_{\nu]} = 2\Big(\frac{\partial_\mu A_\nu - \partial_\nu A_\mu}{2!}\Big)$$

Notice that S_M has a certain gauge symmetry, where the following shift leaves the action invariant:

$$A_\mu \to A_\mu + \partial_\mu \Lambda$$

There are typical choices for Λ. Two are the Coulomb ($\nabla \cdot \vec{A} = 0$) and the Lorentz gauge ($\partial_\mu A^\mu = 0$). From these, we get the Maxwell's equations (equations of motion)

$$\begin{aligned} 0 =& \frac{\partial \mathcal{L}_M}{\partial A_\mu} - \partial_\nu\Big(\frac{\partial \mathcal{L}_\mathcal{M}}{\partial(\partial_\nu A_\mu)}\Big) \\ =& - \partial_\nu\Big(- \frac{1}{2}F^{\rho\sigma}\frac{\partial F_{\rho\sigma}}{\partial(\partial_\nu A_\mu)}\Big) \\ =& - \partial_\nu\Big(- \frac{1}{2}F^{\rho\sigma}(\delta^\nu_\rho\delta^\mu_\sigma - \delta^\mu_\rho\delta^\nu_\sigma)\Big) \\ =& - \partial_\nu(-F^{\nu\mu}) \end{aligned}$$

From the gauge symmetry, we have the Bianchi identity, $\partial_{[\mu}F_{\nu\rho]} = 0$ which implies the no-magnetic monopole and Faraday's law, and the equation of motion yield the Gauss and Ampere law.

Proposition. We provide the general solution through the plane wave solution, which is obtainable under the Lorentz gauge $\partial_\mu A^\mu = 0$.

$$A^r_{\mu,\vec{p}}(x) = \epsilon^r_\mu(\vec{p})e^{ip\cdot x}$$

where $r \in \{1, 2\}$ with $p_\mu p^\mu = p^\mu \epsilon^r_\mu(\vec{p}) = 0$. Observe that

$$\partial^\mu A^r_\mu(x) = ip^\mu\epsilon^r_\mu(\vec{p})e^{ip\cdot x} = 0$$

From this, we can look at the deriviative of the field strength to conclude that our identity indeed holds.

$$\partial^\mu F^r_{\mu\nu} = \partial^\mu\partial_\mu A^r_\nu - \partial^\mu\partial_\nu A^r_\mu = -p^\mu p_\mu \epsilon^r_\mu(\vec{p})e^{ip\cdot x} = 0$$

Definition. (Polarisation vectors) There are two physical ones, where the Lorentz gauge kills one degree of freedom.

$$\varepsilon^r_\mu(\vec{p}) = (\varepsilon^r_0, \vec{\varepsilon}^r) = \Big(- \frac{\vec{p}\cdot\vec{\varepsilon}^r}{p^0}, \vec{\varepsilon}^r\Big)$$

where we have used $p^\mu\varepsilon^r_\mu(\vec{p}) = p^0\varepsilon^r_0 + \vec{p}\cdot\vec{\varepsilon}^r = 0$. There is also the residual gauge from the fact that $\varepsilon^r_\mu(\vec{p}) \to \varepsilon^r_\mu(\vec{p}) + \alpha p_\mu$ which kills the other.

$$\varepsilon^r_\mu(\vec{p}) = \Big(-\frac{\vec{p}\cdot\vec{\varepsilon}^r}{p^0} - \alpha p^0, \vec{\varepsilon}^r + \alpha\vec{p}\Big) = \Big(0, \vec{\varepsilon}^r - \frac{\vec{p}\cdot\vec{\varepsilon}^r}{(p^0)^2}\vec{p}\Big)$$

where $\alpha = -\vec{p}\cdot\vec{\varepsilon}^r/(p^0)^2$. Which implies $(0, \vec{\varepsilon}^{1,2}(\vec{p}))$. We often choose the orthonormal ones,

$$\vec{\varepsilon}^r(\vec{p}) \cdot (\vec{\varepsilon}^s(\vec{p}))^* = \delta^{rs}$$

There are two unphysical ones,

$$\begin{cases} \varepsilon^0_\mu(\vec{p}) = n_\mu = (1,0,0,0) \\ \varepsilon^3_\mu(\vec{p}) = (0,\hat{p}) = \dfrac{p_\mu + n_\mu(p\cdot n)}{|p_\mu + n_\mu(p\cdot n)|} \end{cases}$$

These are necessary to describe virtual photons and are useful in analysing quantum electrodynamic scattering processes.

Theorem. The relations between polarisation vectors are as follows

$$\eta^{\mu\nu}(\varepsilon^r_\mu(\vec{p}))^*\varepsilon^s_\nu(\vec{p}) = \eta^{rs}$$

orthonormality and

$$\sum_{r=0}^{3} \eta_{rr}\varepsilon^r_\mu(\vec{p})\varepsilon^s_\nu(\vec{p})^* = \eta_{\mu\nu}$$

completeness. The above gives

$$\sum_{r=1}^{2} \eta_{rr}\varepsilon^r_\mu(\vec{p})(\varepsilon^s_\nu(\vec{p}))^* = \eta_{\mu\nu} - \frac{p_\mu p_\nu + (p_\mu n_\nu + p_\nu n_\mu)(p\cdot n)}{(p\cdot n)^2}$$

Proposition. Finally, we have

$$A_\mu(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}} \sum_{r=1}^{2} \Big(a^r(\vec{p})\varepsilon^r_\mu(\vec{p})e^{ip\cdot x} + a^r(\vec{p})^*\varepsilon^r_\mu(\vec{p})^* e^{-p\cdot x}\Big)$$

and the canonically conjugate fields and Hamiltonian density is

$$\pi_\mu = \frac{\partial\mathcal{L}_{\mathcal{M}}}{\partial(\partial_0 A^\mu)} = F_{0\mu} \qquad \& \qquad \mathcal{H}_M = \pi^\mu\partial_0 A_\mu - \mathcal{L}_M$$

Theorem. (Noether's theorem) Conservation laws in classical field theory are given by Noether's theorem. Consider an infinitesimal transformation,

$$\phi(x) \to \phi'(x) = \phi(x) = \delta\phi(x)$$

where δ is called a continuous symmetry of the action

$$S[\phi] = \int d^4x\, \mathcal{L}(\phi(x), \partial_\mu\phi(x))$$

The consequence of this is that there is suppose to be a conserved (Noether) current. Directly evaluating $\delta\mathcal{L}$,

$$\begin{aligned}\delta\mathcal{L} =& \frac{\partial\mathcal{L}}{\partial\phi}\,\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\mu(\delta\phi) \\ =& \Big(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\Big)\,\delta\phi + \partial_\mu\Big(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\delta\phi\Big)\end{aligned}$$

Invariance of the action for a continuous symmetry ($\delta S = 0$) is dictated by the condition

$$\delta\mathcal{L} = \partial_\mu f^\mu$$

where $f^\mu \to 0$ fast enough at infinity. Identifying the above, we have, by defining

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\delta\phi - f^\mu$$

the following

$$\partial_\mu j^\mu = 0$$

on-shell. The Noether current above is conserved on-shell! What is the conserved quantity?

$$Q = \int d^3\vec{x}\, j^0$$

as

$$\frac{dQ}{dt} = \int d^3x\, \partial_0 j^0 = \int d^3\vec{x}\, \nabla\cdot\vec{j} = -\int d\vec{a}\cdot\vec{j} = 0$$

There are mutliple types of symmetries. (1) Spacetime symmetries are symmetries that involve transformations of spacetime coordinates. There are also (2) internal symmetries, which are symmetries that act on fields but do not affect spacetime symmetries themselves. For these symmetries, we have $U(1)$, $SU(N)$ symmetries and etcetera.

Lecture 6 (September 17th)

Example. ($U(1)$ symmetry in a complex scalar field theory) Consider the Lagrangian

$$\mathcal{L} = -\partial^\mu\phi\partial_\mu\phi^* - \mathcal{V}(|\phi|^2)$$

The $U(1)$ symmetry is given as

$$\phi'(x) = e^{i\alpha}\phi(x) = \phi(x) + \underbrace{i\alpha\phi(x)}_{\delta\phi(x)} + O(\alpha^2)$$

Invariance of the action is given by $\delta\mathcal{L} = \partial_\mu f^\mu = 0 \to f^\mu = 0$. The Noether current is

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)}\,\delta\phi^* - f^\mu = (-\partial^\mu\phi^*)(i\alpha\phi) + (-\partial^\mu\phi)(-i\alpha\phi^*)$$
$$=\alpha(-i\phi\partial^\mu\phi^* + i\phi^*\partial^\mu\phi)$$

which implies that $\partial_\mu j^\mu = 0$ on-shell. Then, Noether charge would be, physically, the charge carried by particles described by ϕ.

Example. ($U(1)$ symmetries in Dirac theory) Consider the Lagragian

$$\mathcal{L} = \bar{\psi}(i\partial\!\!\!/ - m)\psi$$

The $U(1)$ symmetry is given as

$$\psi'(x) = e^{i\alpha}\psi(x) = \psi(x) + \underbrace{i\alpha\psi(x)}_{\delta\psi(x)} + O(\alpha^2)$$

Again, $\delta\mathcal{L} = 0$ implies that $f^\mu = 0$ and the Noether current is given by

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\,\delta\psi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi^\dagger)}\,\delta\psi^\dagger - f^\mu = i\bar{\psi}\gamma^\mu(i\alpha\psi)$$

and $\partial_\mu j^\mu = 0$. The rather annoying sign can be eliminated via renormalisation giving

$$j^\mu = \bar{\psi}\gamma^\mu\psi$$

Example. (Chiral transformation) Another $U(1)$ symmetry arises for the massless case $m = 0$.

$$\psi'(x) = e^{i\alpha\gamma^5}\psi(x) = \psi(x) + i\alpha\gamma^5\psi(x) + O(\alpha^2)$$

The Nother current is given as, through renormalising $\delta\psi = -i\gamma^5\psi$

$$j_5^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi$$

We call j^μ and j_5^μ are called vectors and axial-vector current respectively. For ψ coupled to an electromagnetic field (EM) A_μ, j^μ is called the electric current density. For a massless ψ coupled with a EM field,

$$\begin{cases} j_L^\mu = \dfrac{1}{2}(j^\mu - j_5^\mu) = \bar{\psi}\gamma^\mu\dfrac{1-\gamma^5}{2}\psi = \bar{\psi}_L\gamma^\mu\psi_L \\ j_R^\mu = \dfrac{1}{2}(j^\mu + j_5^\mu) = \bar{\psi}\gamma^\mu\dfrac{1+\gamma^5}{2}\psi = \bar{\psi}_R\gamma^\mu\psi_R \end{cases}$$

and these denote the electric current densities for left-handed and right-handed Weyl spinors.

Example. (Translation symmetry in a real scalar field) Consider the Lagrangian

$$\mathcal{L} = -\frac{1}{2}\partial^\mu\phi\partial_\mu\phi - \mathcal{V}(\phi)$$

We need to see what $\delta\phi$ is under $x \to x' = x + a$. By definition,

$$\phi'(x') = \phi(x) = \phi(x' - a) = \phi(x') - \underbrace{a^\mu\partial_\mu\phi(x')}_{\delta\phi(x')} + O(a^2)$$

Now what is f^μ? We can easily expect $\delta\mathcal{L} = \mathcal{L}' - \mathcal{L} = -a^\mu\partial_\mu\mathcal{L}$. Therefore, $f^\mu = -a^\mu\mathcal{L}$. Our Noether current is

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi - f^\mu = a_\nu(\partial^\mu\partial^\nu\phi + \eta^{\mu\nu}\mathcal{L})$$

The rank 2 tensor in the second term is called the stress tensor $T^{\mu\nu}$. We find that

$$\partial_\mu j^\mu = 0 = \partial_\mu(a_\nu T^{\mu\nu})$$

for all a^μ and

$$\partial_\mu T^{\mu\nu} = 0$$

It is natural to think that the conserved quantity has four degrees of freedom as we have translated with four degrees of freedom. The Noether charges exactly the 4-momentum!

$$P^\nu = \int d^3\vec{x}\, T^{0\nu}$$

We can then check that

$$P^0 = H = \int d^3\vec{x}(\pi\dot{\phi} - \mathcal{L})$$

Simply beautiful!

Example. (Lorentz symmetry in a real scalar field theory) Consider the infinitesimal transformation $x' = \Lambda x = (1 + \omega)x$. Componentwise, we can write

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu = (\delta^\mu{}_\nu + \omega^\mu{}_\nu)x^\nu$$

Using the Taylor expansion again,

$$\phi'(x') = \phi(x) = \phi(\Lambda^{-1}x') = \phi(x') - \omega^\mu{}_\nu x'^\nu\partial_\mu\phi(x') + O(\omega^2)$$

the trick is that $(1 + \omega)(1 - \omega) = 1 + O(\omega^2)$. What is f^μ?

$$\delta\mathcal{L} = \mathcal{L}' - \mathcal{L} = -\omega^\mu{}_\nu x^\nu\partial_\mu\mathcal{L} = \partial_\mu(-\omega^\mu{}_\nu x^\nu\mathcal{L}) = \partial_\mu f^\mu$$

Thus, $f^\mu = -\omega^\mu{}_\nu x^\nu \mathcal{L}$. The Noether current is correspondingly

$$\begin{aligned}
j^\mu =& \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi - f^\mu \\
=& \omega^{\nu\rho} x_\rho \partial_\nu \phi \partial^\mu \phi + \omega^{\nu\rho} x_\rho \delta^\mu{}_\nu \mathcal{L} \\
=& \omega_{\nu\rho} x^\rho \underbrace{(\partial^\nu \phi \partial^\mu \phi + \eta^{\mu\nu}\mathcal{L})}_{T^{\mu\nu}} \\
=& \frac{1}{2}\omega_{\nu\rho}(x^\rho T^{\mu\nu} - x^\nu T^{\mu\rho})
\end{aligned}$$

we call the second term $\mathcal{J}^{\mu\nu\rho}$. Concluding,

$$\partial_\mu j^\mu = 0$$

for all $\omega_{\nu\rho}$ implies that

$$\partial_\mu \mathcal{J}^{\mu\nu\rho} = 0$$

Thus, the Noether charges are generalised (with boosts) angular momenta!

$$J^{\mu\nu} = \int d^3\vec{x}\, \mathcal{J}^{0\mu\nu}$$

for example, $\mu\nu = 12$ shows

$$\int d^3\vec{x}\, (x^2 T^{01} - x^1 T^{02})$$

Remark. There are also discrete symmetries that we learn from here.

Definition. (Space inversion or parity) The transformation is given by

$$x = (t, \vec{x}) = (t', \vec{x}') = (t, -\vec{x})$$

The parity on fields give

$$\phi'(x') = \phi(x) \qquad \& \qquad \phi'(x') = \gamma^0 \psi(x)$$

lastly,

$$A'^\mu(x') = \begin{cases} A^\mu(x) & (\mu = 0) \\ -A^\mu(x) & (\mu = 1, 2, 3) \end{cases}$$

Example. (Parity on $U(1)$ currents in Dirac theory) For vectors,

$$j^\mu = \bar{\psi}\gamma^\mu\psi \to (\gamma^0\psi)^\dagger \gamma^0 \gamma^\mu (\gamma^0 \psi) = \psi^\dagger \gamma^\mu \gamma^0 \psi$$

and

$$\begin{cases} \bar{\psi}\gamma^0\psi & (\mu = 0) \\ -\bar{\psi}\gamma^i\psi & (\mu = i) \end{cases}$$

while for axial-vectors,

$$j_5^\mu = \bar{\psi}\gamma^\mu\gamma_5\psi \to (\gamma^0\psi)^\dagger\gamma^0\gamma^\mu\gamma_5(\gamma^0\psi) = \psi^\dagger\gamma^\mu\gamma_5\gamma^0\psi$$

and

$$\begin{cases} -\bar{\psi}\gamma^0\gamma_5\psi & (\mu = 0) \\ \bar{\psi}\gamma^\mu\gamma_5\psi & (\mu = i) \end{cases}$$

Definition. (Charge conjugation) This transformation simply swaps particles with anti-particles and has nothing to do with position. Charge conjugation on fields result in

$$\phi'(x) = \phi(x) \& \phi'(x) = C\gamma^0\psi^*(x) \& A'_\mu(x) = -A_\mu(x)$$

here,

$$C\gamma^\mu C^{-1} = -(\gamma^\mu)^T \& C^{-1} = C^\dagger = C^T = -C$$

and a typical choice is

$$C = i\gamma^2\gamma^0$$

Definition. (Time reversal) The time reversal transformation of x is given as $x = (t, \vec{x}) \to (t', \vec{x}') = (-t, \vec{x})$. The time reversal on fields give

$$\phi'(x') = \phi(x)^* \qquad \& \qquad \psi'(x') = T\psi(x)^*$$

and finally

$$A'^\mu(x') = \begin{cases} A^\mu(x) & (\mu = 0) \\ -A^\mu(x) & (\mu = i) \end{cases}$$

Here, T is the time-reversal matrix, satisfying $T^{-1}\gamma^\mu T = (\gamma^\mu)^T \quad \& \quad T = T^{-1} = T^\dagger = -T^* = -T^T$ again, a typical choice is

$$T = i\gamma^1\gamma^3 = -\gamma^5 C$$

Lecture 7 (September 22nd)

Free Quantum Field Theory

Remark. There are two different approaches to quantum field theory.

(i) (Canonical quantisation) Useful to see how classical field theories become free quantum field theories. This is analogous to second quantisation, and lets us understand the resulting $\mathcal{H}$ more explicitly.

(ii) (Path integral) Useful to studying interacting quantum field theories. You'll be seeing a lot of expressions like

$$\langle 0|T\hat{\phi}(x_1)\dots\hat{\phi}(x_n)|0\rangle$$

involving time ordered correlated functions. This is much easier to study in path integrals.

Definition. (Canonical quantisation) The standard dogma from classical field theory into quantum field theory is

(i) All dynamical fields become operators

$$(\phi, \pi, \psi, \pi_\psi) \mapsto (\hat{\phi}, \hat{\pi}, \hat{\psi}, \hat{\pi}_\psi)$$

(ii) Poisson brackets become commutator relations

$$\{\cdot,\cdot\}_{PB} \mapsto \frac{1}{i\hbar}[\cdot,\cdot]$$

(note that from now on, we will be using natural units with $\hbar = 1$)

Remark. We make the following remarks:

(i) Most books follow the Heisenberg picture, with operators evolving over time and states remaining the same.

(ii) To incorporate fermions, the Poisson brackets become super-poisson brackets and commutators become anti-commutators.

(iii) The spin-statistics theorem dictates the quantisation rules.

(iv) 2nd quantisation is very different from 1st quantisation in quantum mechanics.

(v) We will use the abbreviation ETCR for equal time correlation relations.

Theorem. (Spin-0 Klein-Gordan Fields) We first consider the classical Klein-Gordan action,

$$S_{KG} = \int d^4x \left(-\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 \right)$$

After canonical quantisation (ETCR) the fields $(\phi, \pi) = (\phi, \dot{\phi})$ become operators and we impose that

$$\begin{cases} [\phi(t,\vec{x}), \phi(t,\vec{y})] = i\delta^{(3)}(\vec{x}-\vec{y}) \\ [\phi(t,\vec{x}), \phi(t,\vec{y})] = [\pi(t,\vec{x}), \pi(t,\vec{y})] = 0 \end{cases}$$

What about the equations of motion? Let's first observe that the Hamiltonian becomes

$$\hat{H} = \int d^3\vec{x}\, \frac{1}{2}(\hat{\pi}^2 + (\nabla\hat{\phi})^2 + m^2\hat{\phi}^2)$$

Hamilton's equations then become

$$\frac{d\hat{\phi}}{dt} = -i[\hat{\phi}, \hat{H}] \qquad \& \qquad \frac{d\hat{\pi}}{dt} = -i[\hat{\pi}, \hat{H}]$$

we can then show that they automatically follow

$$(\partial^\mu \partial_\mu - m^2)\hat{\phi} = 0$$

Soltuions to the quantised Klein-Gordan equation become

$$\phi(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 \cdot 2E_p}} \left(\hat{a}(\vec{p}) e^{ip\cdot x} + \hat{a}(\vec{p})^\dagger e^{-ip\cdot x} \right)$$

The quantised conjugate field is then

$$\hat{\pi}(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 \cdot 2E_p}} (-iE_p) \Big(\hat{a}(\vec{p}) e^{ip\cdot x} - \hat{a}(\vec{p})^\dagger e^{-p\cdot x} \Big)$$

The inverse relations are also given as

$$u_{\vec{p}}(x) = \frac{1}{\sqrt{(2\pi)^3 \cdot 2E_p}} e^{ip\cdot x} \qquad \& \qquad A \overset{\leftrightarrow}{\partial} B = A(\partial B) - (\partial A) B$$

Then,

$$\begin{cases} \hat{a}(\vec{p}) = i \int d^3x \, u^*_{\vec{p}} \partial_0^{\leftrightarrow} \hat{\phi}(x) \\ \hat{a}(\vec{p})^\dagger = -i \int d^3\vec{x} \, u_{\vec{p}}(x) \, \partial_0^{\leftrightarrow} \hat{\phi}(x) \end{cases}$$

Using

$$\int \frac{d^3\vec{x}}{(2\pi)^3} \, e^{i\vec{p}\cdot\vec{x}} = \delta^{(3)}(\vec{p})$$

The commmutation relations for the Fourier mode operators are given as

$$[\hat{a}(\vec{p}), \hat{a}(\vec{q})^\dagger] = \delta^{(3)}(\vec{p} - \vec{q}) \qquad \& \qquad [\hat{a}(\vec{p}), \hat{a}(\vec{q})] = [\hat{a}(\vec{p})^\dagger, \hat{a}(\vec{q})^\dagger] = 0$$

This can be seen as infinitely many sets of creation and annihilation operators for a harmonic oscillator.

Proposition. (Fock space $\mathcal{H}$) Given the above, we want to now construct the Hilbert space that these fields should act on using the ladder operator method. We define

(i) For all $\vec{p}$,

$$\hat{a}(\vec{p})|0\rangle = 0$$

(ii) A single particle state with momentum $\vec{p}$

$$|\vec{p}\rangle = \hat{a}(p)^\dagger |0\rangle$$

(iii) For a multiple particle state (for example a two particle state with momentum $\vec{p}$ and $\vec{q}$),

$$|\vec{p},\vec{q}\rangle = \hat{a}(\vec{p})^\dagger \hat{a}(\vec{q})^\dagger |0\rangle$$

Therefore, we can think that the Fock space is a vector space made out of all linear combinations of all possible multipe particle states.

Definition. (Momentum and Hamiltonian) The momentum operator is given as

$$\hat{P}^\mu = \int d^3\vec{x}\,\hat{T}^{0\mu} = \int d^3\vec{x}\Big(\partial^0\hat{\phi}\partial^\mu\hat{\phi} + \eta^{0\mu}\Big(-\frac{1}{2}(\partial\hat{\phi})^2 - \frac{1}{2}m^2\hat{\phi}^2\Big)\Big)$$

whereas the Hamiltonian operator is given as

$$\begin{aligned}
\hat{H} =& \frac{1}{2}\int d^3\vec{x}\,(\hat{\pi}^2 + (\nabla\phi)^2 + m^2\hat{\phi}^2)\\
=& \frac{1}{2}\int d^3\vec{x}\int \frac{d^3\vec{p}\,d^3\vec{q}}{(2\pi)^3\sqrt{2E_p\cdot 2E_p}}\Big[(-iE_p)(-iE_q)\Big(\hat{a}_p e^{ip\cdot x} - \hat{a}_p^\dagger e^{-ip\cdot x}\Big)\Big(\hat{a}_q e^{ipq\cdot x} - \hat{a}_q^\dagger e^{-iq\cdot x}\Big)\Big] + \dots\\
=& \frac{1}{2}\int \frac{d^3\vec{p}}{2E_p}\Big((\hat{a}_p\hat{a}_{-p}e^{-2iE_pt} + \hat{a}_p^\dagger\hat{a}_{-p}^\dagger e^{2iE_pt})(-E_p^2 + \vec{p}^2 + m^2) + (\hat{a}_p\hat{a}_p^\dagger + \hat{a}_p^\dagger\hat{a}_p)(E_p^2 + \vec{p}^2 + m^2)\Big)\\
=& \frac{1}{2}\int d^3\vec{p}\,E_p(\hat{a}^\dagger\hat{a}(\vec{p}) + \hat{a}(\vec{p})\hat{a}(\vec{p})^\dagger)\\
=& \int d^3\vec{p}\,E_p\hat{a}(\vec{p})^\dagger\hat{a}(\vec{p}) + \int d^3\vec{p}\,\frac{E}{2}\delta^{(3)}(0)
\end{aligned}$$

Doing something similar to momentum,

$$\begin{aligned}
\hat{\vec{p}} =& \int d^3x\,(-\hat{\pi}\nabla\hat{\phi})\\
=& \int d^3\vec{p}\,\vec{p}\hat{a}^\dagger(\vec{p})\hat{a}(\vec{p}) + \frac{1}{2}\int d^3\vec{p}\,\vec{p}\delta^{(3)}(\vec{0})
\end{aligned}$$

where the second term is named the vacuum momentum $\vec{P}_0$. We have some comments:

(i) Vaccum energy can be understood as the sum of all ground state energies for harmonic oscillators and is divergent!

Lecture 8 (September 24th)

Remark. (I) The vacuum energy E is the sum of all ground state energies for the harmonic oscillator, and therefore is divergent! We might think of possible resolutions.

(i) Ignore, as energy is only meanful as a difference

(ii) Introducing some regulator called the IR regulator, taking

$$\delta^{(3)}(\vec{p}) = \int \frac{d^3\vec{x}}{(2\pi)^3}\, e^{i\vec{p}\cdot\vec{x}} \longrightarrow \delta^{(3)}(\mathbf{0}) = \int \frac{d^3\vec{x}}{(2\pi)^3} = \frac{V}{(2\pi)^3}$$

which implies that

$$E_0 = \frac{V}{(2\pi)^3} = \int d^3\vec{p}\,(E_p/2)$$

(iii) Renormalisation, and redefining our theory by employing a infinite counterterm

$$\mathcal{L}_{ren} = -\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m\phi^2 + \Lambda_0 \longrightarrow \hat{H}_{rem} = \hat{H} - \int d^3\vec{x}\,\Lambda_0$$

where Λ_0 would be the cosmological constant and we would set it such that the integral in the right would become E_0. In this case, the action would become infinite, but the observables would become finite!

(II) Another important remark is that the order of the operators matter. We define the normal ordering to be putting creation operators to the left. Well, how could this be helpful?

(III) The Klein-Gordan field operators satisfy the time evolution property in the Heisenberg picture. Well, see that

$$\begin{aligned}
e^{i\hat{H}x^0}\hat{\phi}(0,\vec{x})e^{-i\hat{H}x^0} &= \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}} e^{i\hat{H}x^0}\left(\hat{a}(\vec{p})e^{i\vec{p}\cdot\vec{x}} + \hat{a}(\vec{p})^\dagger e^{-i\vec{p}\cdot\vec{x}}\right)e^{-i\hat{H}x^0}\\
&= \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}}\left(\hat{a}(\vec{p})e^{-iE_px^0+i\vec{p}\cdot\vec{x}} + \hat{a}(\vec{p})^\dagger e^{iE_px^0-i\vec{p}\cdot\vec{x}}\right)\\
&= \hat{\phi}(x^0,\vec{x}) = \hat{\phi}(x)
\end{aligned}$$

with

$$\begin{aligned}
[\hat{H},\hat{a}(\vec{p})] &= \int d^3\vec{q}\, E_q[\hat{a}_q^\dagger\hat{a}_q,\hat{a}_p]\\
&= \int d^3\vec{q}\, E_q\left(\hat{a}_q^\dagger[\hat{a}_q,\hat{a}_p] + [\hat{a}_p^\dagger,\hat{a}_p]\hat{a}_q\right)\\
&= -E_p\hat{a}(\vec{p})
\end{aligned}$$

and

$$[\hat{H},\hat{a}(\vec{p})^\dagger] = E_p\hat{a}(\vec{p})^\dagger$$

Notably, the commutator relations with the ladder operators and the Hamiltonian allows the fields to transform unitarily under the time-evoluation operator. For general spacetime translations, we'll see also that

$$e^{-i\hat{P}_\mu a^\mu}\hat{\phi}(x)e^{i\hat{P}_\mu a^\mu} = \hat{\phi}(x+a)$$

this characterises the translation invariance of a scalar field operator.

Remark. (Testing the Fock space) Acting the time-ordered Hamiltonian on the momentum ket, we find

$$\hat{H}|\vec{p}\rangle = \int d^3\vec{q}\, E_q \hat{a}_q^\dagger [\hat{a}_q, \hat{a}_p^\dagger]|0\rangle = E_p \hat{a}_p^\dagger|0\rangle = E_p|\vec{p}\rangle$$

and

$$\hat{H}|\vec{p}_1, \ldots, \vec{p}_N\rangle = (E_{p_1} + \ldots E_{p_n})|\vec{p}_1, \ldots, \vec{p}_N\rangle$$

which is the intuition we have. Also,

$$\hat{P}|\vec{p}_1, \ldots, \vec{p}_n\rangle$$

We also make the note that if we deal with complex Klein-Gordan theory, we get another interesting charge which is associated with the $U(1)$ symmetry!

Definition. (Feynman propagator and the time ordered product) The Feynman propagator is a crucial ingredient in interacting quantum field theory.

$$\Delta_F(x, y) = \langle 0|T\hat{\phi}(x)\hat{\phi}(y)|0\rangle = \Delta_F(x - y)$$

where the equality followings from translation symmetry. The T that we wrote is the time ordered product defined as

$$T\hat{A}(x)\hat{B}(y) = \Theta(x^0 - y^0)\hat{A}(x)\hat{B}(y) \pm \Theta(y^0 - x^0)\hat{B}(y)\hat{A}(x)$$

where the plus is for bosonic cases and the minus is for fermionic cases. Let's try evaluating the Feynman propagator. Having

$$\hat{\phi}(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}}\left(\hat{a}_p e^{ip\cdot x} + \hat{a}_p^\dagger e^{-ip\cdot x}\right) \qquad \& \qquad \langle 0|\hat{a}_p^\dagger = \hat{a}_p|0\rangle = 0$$

This then gives

$$\begin{aligned}
\Delta_F(x - y) &= \int \frac{d^3\vec{p}d^3\vec{q}}{(2\pi)^3\sqrt{2E_p 2E_q}}\Big(\Theta(x^0 - y^0)\langle 0|[\hat{a}_q, \hat{a}_q^\dagger]|0\rangle e^{i(p\cdot x - q\cdot y)} \\
&\quad + \Theta(y^0 - x^0)\langle 0|[\hat{a}_q, \hat{a}_p^\dagger]|0\rangle e^{-i(p\cdot x - q\cdot y)} \\
&= \int \frac{d^3\vec{p}}{(2\pi)^3 2E_p}\left(\Theta(x^0 - y^0)e^{ip\cdot(x-y)} + \Theta(y^0 - x^0)e^{-ip\cdot(x-y)}\right) \\
&= \int \frac{d^3\vec{p}}{(2\pi)^3}e^{i\vec{p}\cdot(\vec{x}-\vec{y})} \times \frac{1}{2E_p}\left(\Theta(x^0 - y^0)e^{-iE_p(x^0-y^0)} + \Theta(y^0 - x^0)e^{-iE_p(x^0-y^0)}\right) \\
&= \int \frac{d^4p}{(2\pi)^4}\frac{-ie^{ip\cdot(x-y)}}{p^2 + m^2 - i\varepsilon}
\end{aligned}$$

where $\varepsilon\varepsilon_0 > 0$ is arbitrarily small. Surprisingly,

$$\int \frac{dp^0}{2\pi i}\frac{e^{-ip^0(x^0-y^0)}}{-(p^0)^2+\vec{p}^2+m^2-i\varepsilon} = \int \frac{dp^0}{2\pi i}\frac{-e^{-ip^0(x^0-y^0)}}{(p^0+E_p-i\varepsilon)(p^0-E_p+i\varepsilon)}$$

with two residues at $-E_p$ and E_p each. We find

$$\Theta(x^0-y^0)\Big(\frac{e^{-iE_p(x^0-y^0)}}{2E_p}\Big)+\Theta(y^0-x^0)\Big(\frac{-e^{iE_p(x^0-y^0)}}{-2E_p}\Big)$$

Proposition. The Feynman propagator acts as a Green's function of the Klein-Gordan equation, with

$$\begin{aligned}
&(-\partial_x^2+m^2-i\varepsilon)\Delta_F(x-y)\\
&\qquad\qquad =(-\partial_x^2+m^2-i\varepsilon)\int\frac{d^4p}{(2\pi)^4}\cdot\frac{-ie^{ip\cdot(x-y)}}{p^2+m^2-i\varepsilon}\\
&\qquad\qquad =\int\frac{d^4p}{(2\pi)^4}\,(p^2+m^2-i\varepsilon)\frac{-ie^{ip\cdot(x-y)}}{p^2+m^2-i\varepsilon}=-i\delta^{(4)}(x-y)
\end{aligned}$$

where $p^\mu p_\mu = p^2$.

Lecture 9 (September 29th)

Definition. (Spin-$\frac{1}{2}$ Dirac Fields) The Dirac action was given as

$$S_D = \int d^4x\,\bar{\psi}(i\partial\!\!\!/-m)\psi$$

The canonical quantisation or ETCR were give as

$$\hat{\psi}(x),\hat{\pi}(x)=i\hat{\psi}^\dagger(x)\longrightarrow\begin{cases}\{\hat{\psi}_\alpha(t,\vec{x}),\hat{\psi}_\beta(t,\vec{y})\}=\{\hat{\pi}_\alpha(t,\vec{x}),\hat{\pi}_\beta(t,\vec{y})\}=0\\ \{\hat{\psi}_\alpha(t,\vec{x}),\hat{\pi}_\beta(t,\vec{y})\}=i\delta^{(4)}_{\alpha\beta}(\vec{x}-\vec{y})\end{cases}$$

where the symmetriser comes from the spin-statistics theorem (anti-commutation for fermions). The equations of motion are promoted from the Hamiltonian as

$$\hat{H}=\int d^3\vec{x}\,\hat{\psi}^\dagger(-i\gamma^0\vec{\gamma}\cdot\vec{\nabla}+\gamma^0 m)\hat{\psi}$$

The Hamilton's equation become

$$\frac{d\hat{\psi}}{dt}=-i[\hat{\psi},\hat{H}]\longrightarrow(-i\partial\!\!\!/-m)\hat{\psi}=0$$

The solutions to the quantised Dirac equation look like

$$\hat{\psi}(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}} \sum_{i=1}^{2} \left(\hat{b}^i(\vec{p}) u_+^i(\vec{p}) e^{ip\cdot x} + \hat{d}_{\vec{p}}^{i\,*} u_-^i(\vec{p}) e^{-ip\cdot x} \right)$$

while the conjugate field looks like

$$\hat{\pi}(x) = i \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}} \sum_{i=1}^{2} \left(\hat{b}^i(\vec{p})^\dagger u_+^i(\vec{p})^\dagger e^{-ip\cdot x} + \hat{d}^i(\vec{p}) u_-^i(\vec{p})^\dagger e^{ip\cdot x} \right)$$

Recall that after we found the fields, we wanted the mode operators in terms of the fields and we obtain

$$\hat{b}^i(\vec{p}) = \int \frac{d^3\vec{x}}{\sqrt{(2\pi)^3 2E_p}} \, u_+^i(\vec{p})^\dagger \hat{\psi}(x) e^{-ip\cdot x}$$

and

$$\hat{d}^i(\vec{p})^\dagger = \int \frac{d^3\vec{x}}{\sqrt{(2\pi)^3 2E_p}} \, u_-^i(\vec{p})^\dagger \hat{\psi}(x) e^{ip\cdot x}$$

(this can be proved through using $u_\pm^i(\vec{p})^\dagger u_\pm^j(\vec{p}) = 2E_p \delta^{ij}$ and $u_\pm^i(\vec{p}) u_\mp^j(-\vec{p}) = 0$). The anti-commuation relations for $\hat{b}^i$ and $\hat{d}^i$ should be

$$\{\hat{b}^i(\vec{p}), \hat{b}^j(\vec{q})^\dagger\} = \{\hat{d}^i(\vec{p}), \hat{d}^j(\vec{q})^\dagger\} = \delta^{ij} \delta^{(3)}(\vec{p} - \vec{q})$$

which is exactly the infinitely many sets of creation and annihilation operators for a fermionic harmonic oscillator.

Definition. (Construction of the Hilbert/Fock space) The vaccum state is defined using the annihilator operator

$$\hat{b}^i(\vec{p})|0\rangle = \hat{d}^{i'}(\vec{p})|0\rangle = 0$$

The first excited states look something like

$$\hat{b}^i(\vec{p})^\dagger|0\rangle = |(\vec{p}, i)\rangle \qquad \& \qquad \hat{d}^{i'}(\vec{p})^\dagger|0\rangle = |(\vec{p}', i')\rangle$$

The multiple particle states trivially follow as

$$|(\vec{p}, i_1), \ldots, (\vec{p}_M, i_N), (\vec{p}_1', i_1'), \ldots, (\vec{p}_N', i_N')\rangle = \hat{b}^{i_1}(\vec{p}_1)^\dagger \ldots \hat{d}^{i_N'}(\vec{p}_N')^\dagger|0\rangle$$

Theorem. (Pauli exclusion principle) This naturally follows from

$$\left(\hat{b}^i(\vec{p})^\dagger\right)^2 = \left(\hat{d}^{i'}(\vec{p}')\right)^2 = 0$$

and the general multiple particle state vanishes when exists (m, n) such that

$$(\vec{p}_m, i_m) = (\vec{p}_n, i_n) \qquad \text{or} \qquad (\vec{p}_m{}', i_m') = (\vec{p}_n{}', i_n')$$

Definition. The momentum and Hamiltonian in terms of the mode operators are

$$\hat{P}^\mu = \int d^3\vec{x}\,\hat{T}^{0\mu} = -\int d^3\vec{x}\, i\hat{\bar{\psi}}\gamma^0\partial^\mu\hat{\psi}$$

and

$$\hat{H} = i\int d^3\vec{x}\,\hat{\psi}^\dagger\partial_t\hat{\psi}$$

resulting in

$$\begin{aligned}
\hat{H} =& i\int \frac{d^3\vec{p}}{2E_p}(-iE_p)\sum_{i,j=1}^{2} 2E_p\delta^{ij}\left(\hat{b}^i(\vec{p})^\dagger\hat{b}^j(\vec{p}) - \hat{d}^i(\vec{p})\hat{d}^j(\vec{p})^\dagger\right)\\
=& \int d^3\vec{p}\,E_p\sum_{i=1}^{2}\left(\hat{b}^i(\vec{p})^\dagger\hat{b}^i(\vec{p}) - \hat{d}^i(\vec{p})\hat{d}^i(\vec{p})^\dagger\right)\\
=& \int d^3\vec{p}\,E_p\sum_{i=1}^{2}\left(\hat{b}^i(\vec{p})^\dagger\hat{b}^i(\vec{p}) + \hat{d}^i(\vec{p})^\dagger\hat{d}^i(\vec{p})\right) - \int d^3\vec{p}\,E_p\delta^{(3)}(\mathbf{0})
\end{aligned}$$

The momentum operator is naturally

$$\begin{aligned}
\hat{\vec{P}} =& -i\int d^3\vec{x}\,\hat{\psi}^\dagger\nabla\hat{\psi}\\
=& \int d^3\vec{p}\,\vec{p}\sum_{i=1}^{2}\left(\hat{b}^i(\vec{p})^\dagger\hat{b}^i(\vec{p}) + \hat{d}^i(\vec{p})^\dagger\hat{d}^i(\vec{p})\right)
\end{aligned}$$

There are also other important conserved charges! The $U(1)$ symmetry becomes the particle numbers ($+1$ for $\hat{b}^\dagger$ and -1 for $\hat{d}^\dagger$). The Noether current is $j^\mu = \bar{\psi}\gamma^\mu\psi$ leading to the conserved quantity

$$\hat{N} = \int d^3\vec{x}\,\hat{j}^0 = \int d^3\vec{x}\,\hat{\psi}^\dagger\hat{\psi}$$

and

$$\begin{aligned}
\hat{N} =& \int d^3\vec{p}\sum_{i=1}^{2}\left(\hat{b}^i(\vec{p})^\dagger\hat{b}^i(\vec{p}) + \hat{d}^i(\vec{p})\hat{d}^i(\vec{p})^\dagger\right)\\
=& \int d^3\vec{p}\sum_{i=1}^{2}\left(\hat{b}^i(\vec{p})^\dagger\hat{b}^i(\vec{p}) - \hat{d}^i(\vec{p})^\dagger\hat{d}^i(\vec{p})\right) + N_0
\end{aligned}$$

Also, the normal Lorentz symmetries give the generalised angular momenta. For the Dirac field,

$$\mathcal{J}^{\mu\nu\rho} = \underbrace{x^\rho T^\mu - x^\nu T^{\mu\rho}}_{\text{orbital}} + \underbrace{\bar{\psi}\gamma^\mu\left(\frac{i}{2}\gamma^{\nu\rho}\right)\psi}_{\text{spinorial}=\mathcal{S}^{\mu\nu\rho}}$$

We are compelled to think that

$$\begin{cases} \hat{J}^{\mu\nu} = \int d^3\vec{x}\, \hat{\mathcal{J}}^{0\mu\nu} \\ \hat{S}^{\mu\nu} = \int d^3\vec{x}\, \hat{\mathcal{S}}^{0\mu\nu} \end{cases}$$

is the second part generalised spin? There are a lot of subtleties, addressed in the homework problems.

Remark. The vaccum energy E can be seen as the sum of all ground state energies of the fermionic oscillators. They are therefore negative and divergent. Let us interpret the ground state recalling that

$$\{\hat{d}, \hat{d}^\dagger\} \qquad \& \qquad \hat{d}|0\rangle = 0$$

where the first is symmetric.

The first perspective is that $\hat{d}^\dagger$ and $\hat{d}$ create and annihilate positive energy states. Therefore, $|0\rangle$ can be seen as the vaccum without any excited positive energy states. Also, $\hat{b}^\dagger$ and $\hat{d}^\dagger$ would be creating particles and anti-particle states (positive).

The second perspective is that $\hat{d}$ and $\hat{d}^\dagger$ create and annihilate negative energy states. Then, the state $|0\rangle$ would be interpreted as a state with fully filled negative energy states, exactly the "Dirac sea". Also, $\hat{d}^\dagger$ would create a hole in the Dirac sea.

Definition. (Normal ordering) Again, the normal ordering operator sends all annihilation operators to the right, giving

$$:\hat{H} := \int d^3\vec{p}\, E_p \sum_{i=1}^{2} \Big(\hat{b}^i(\vec{p})^\dagger \hat{b}^i(\vec{p}) + \hat{d}^i(\vec{p})^\dagger \hat{d}^i(\vec{p})\Big)$$

$$:\hat{\vec{P}} := \int d^3\vec{p}\,\vec{p} \sum_{i=1}^{2} (\dots)$$

$$:\hat{N} := \int d^3\vec{p} \sum_{i=1}^{2} \Big(\hat{b}^i(\vec{p})^\dagger \hat{b}^i(\vec{p}) - \hat{d}^i(\vec{p})^\dagger \hat{d}^i(\vec{p})\Big)$$

Note that $:\hat{a}\hat{a}^\dagger := \hat{a}^\dagger\hat{a}$ for bosons whereas $:\hat{b}\hat{b}^\dagger := -\hat{b}\hat{b}$ for fermions.

Lecture 10 (October 1st)

Theorem. Last class we have found the Hamiltonian, momentum operator, and the particle number operator for the Fock space for the Dirac field. We want to now test this Fock space.

$$:\hat{H} : |(\vec{p}, i)\rangle =: \hat{H} : \hat{b}^i(\vec{p})^\dagger |0\rangle = E_p |(\vec{p}, i)\rangle$$

and

$$:\hat{H}:|(\vec{p}_1,i_1),\ldots,(\vec{p}'_N,i'_N)\rangle=(E_{p_1}+\ldots E_{p'_N})|\cdot\rangle$$

The momentum operator acts as

$$:\hat{\vec{P}}:|(\vec{p}_1,i_1),\ldots,(\vec{p}'_N,i'_N)\rangle=(\vec{p}_1+\ldots+\vec{p}'_N)|\cdot\rangle$$

The particle number opeartor

$$:\hat{N}:|(\vec{p}_1,i_1),\ldots(\vec{p}_M,i_M),(\vec{p}'_1,i'_1),\ldots,(\vec{p}'_N,i'_N)\rangle=(M-N)|\cdot\rangle$$

The generalised momentum is rather complicated. Let's focus on the special case! For example, lets take the angular momentum operator acting on a single particle, zero momentum state. We find

$$\hat{S}_z=\int d^3\vec{x}\,\hat{S}^{012}\longrightarrow\hat{S}_z|(\mathbf{0},i)\rangle=\pm\frac{1}{2}|(\mathbf{0},i)\rangle$$

Definition. (Feynman propagator) For the Dirac field, we will be writing

$$S_{F\alpha\beta}(x,y)=\langle 0|T\hat{\psi}_\alpha(x)\hat{\bar{\psi}}_\beta(y)|0\rangle$$

This is equal to

$$S_{F\alpha\beta}(x,y)=\int\frac{d^3\vec{p}\,d^3\vec{q}}{(2\pi)^3\sqrt{2E_p\cdot 2E_q}}\sum_{i,j=1}^{2}\Big[\Theta(x^0-y^0)u^i_{+\alpha}(\vec{p})\bar{u}^j_{+\beta}(\vec{q})\langle 0|\hat{b}^i(\vec{p})\hat{b}^j(\vec{q})^\dagger|0\rangle e^{i(p\cdot x-q\cdot y)}$$
$$-\Theta(y^0-x^0)u^i_{-\alpha}(\vec{p})\bar{u}^j_{-\beta}(\vec{q})\langle 0|\hat{d}^j(\vec{q})\hat{d}^i(\vec{p})^\dagger|0\rangle e^{-i(p\cdot x)-q\cdot y}\Big]$$

$$=\int\frac{d^3\vec{p}}{(2\pi)^3 2E_p}\sum_{i=1}^{2}\Big[\Theta(x^0-y^0)u^i_{+\alpha}(\vec{p})\bar{u}^i_{+\beta}(\vec{p})e^{-ip\cdot(x-y)}-\Theta(y^0-x^0)u^i_{-\alpha}(\vec{p})\bar{u}^i_{-\beta}(\vec{p})e^{-ip\cdot(x-y)}\Big]$$

where we used

$$\sum_{i=1}^{2}u^i_{\pm\alpha}(\vec{p})\bar{u}^i_{\pm\beta}(\vec{p})=(-\not{p}\pm m\mathbf{1}_4)_{\alpha\beta}\qquad\&\qquad\langle 0|\hat{b}^i(\vec{p})\hat{b}^j(\vec{q})^\dagger|0\rangle=\delta^{ij}\delta^{(3)}(\vec{p}-\vec{q})$$

This becomes

$$=\int\frac{d\vec{p}}{(2\pi)^3 2E_p}\Big[\Theta(x^0-y^0)(-\not{p}+m\mathbf{1}_4)_{\alpha\beta}e^{ip\cdot(x-y)}$$
$$+\Theta(y^0-x^0)(\not{p}+m\mathbf{1}_4)_{\alpha\beta}e^{-ip\cdot(x-y)}\Big]$$

$$=(i\not{\partial}_x+m\mathbf{1}_4)_{\alpha\beta}\int\frac{d^3\vec{p}}{(2\pi)^3 2E_p}\Big[\Theta(x^0-y^0)e^{ip\cdot(x-y)}+\Theta(y^0-x^0)e^{ip\cdot(x-y)}\Big]$$
$$=(i\not{\partial}_x+m\mathbf{1}_4)_{\alpha\beta}\Delta_F(x-y)$$

where

$$\Delta_F(x-y) = \int \frac{d^4\vec{p}}{(2\pi)^4} \frac{-i}{p^2+m^2-i\varepsilon} e^{ip\cdot(x-y)}$$

and

$$S_{F\alpha\beta}(x,y) = \int \frac{d^4\vec{p}}{(2\pi)^4} \frac{i(\not{p}-m\mathbf{1}_4)_{\alpha\beta}}{p^2+m^2-i\varepsilon} e^{ip\cdot(x-y)} = \Big[\int \frac{d^4p}{(2\pi)^4} \frac{-i}{\not{p}+m-i\varepsilon} e^{ip\cdot(x-y)}\Big]_{\alpha\beta}$$

Let's now verify that the Feynman propagator is indeed the green function of the Dirac equation.

$$\begin{aligned}(i\not{\partial}_x - m + i\varepsilon)S_F(x-y) =& (-\not{\partial}_x - m + i\varepsilon) \int \frac{d^4p}{(2\pi)^4} \frac{-i}{\not{p}+m-i\varepsilon} e^{ip\cdot(x-y)} \\ =& \int \frac{d^4p}{(2\pi)^4} i e^{ip\cdot(x-y)} \mathbf{1}_4 = i\mathbf{1}_4 \delta^{(4)}(x-y)\end{aligned}$$

Theorem. (Spin-1 Maxwell fields) These have a subtlety due to gauge symmetry. There are esstentially two approaches, (1) non-covariant quantisation and (2) covariant quantisation. For (1),

(i) Based on the Coulomb gauge $\nabla \cdot \vec{A} = 0$.

(ii) Free from unphysical longitudinal degrees of freedom.

(iii) Lacks manifest Lorentz covariance, and is unsuitable for pertubative quantum field theory.

For (2),

(i) Based on the Lorentz gauge $\partial_\mu A^\mu = 0$.

(ii) Unphysical longitudinal mode is quantised. In this case, the associated states are eliminated from the Hilbert space (Gupta-Bleuler formalism).

(iii) Preserves manifest Lorentz invariance and is suitable for pertubative quantum field theory.

Remark. Useful expressions for covariant quantisation

$$\hat{A}_\mu(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}} \sum_{r=0}^{3} \Big(\hat{a}^r(\vec{p})\epsilon^r_\mu(\vec{p})e^{ip\cdot x} + \hat{a}^r(\vec{p})^\dagger \epsilon^r_\mu(\vec{p})^* e^{-ip\cdot x}\Big)$$

Taking

$$u_{\vec{p}}(\vec{x}) = \frac{1}{\sqrt{(2\pi)^3 2E_p}} e^{ip\cdot x}$$

we have

$$\hat{a}^r(\vec{p}) = i\eta^{rr} \int d^3\vec{x}\, \varepsilon^r_\mu(\vec{p})^* u_{\vec{p}}(\vec{x})^* \overleftrightarrow{\partial_0} \hat{A}\mu(x)$$

Here are some comments

(i) $\partial_\mu \hat{A} \neq 0$ (Lorentz gauge does not hold as an operator identity)

(ii) The Gupta-Bleur is used to enforce the Lorentz gauge

Theorem. (Spin-statistics theorem) The spin-statistics theorem states that integer spin particles are bosons, following Bose-Einstein statistics. Then, quantisation is achieved through commutator relations. On the other hand, half-integer spin particles are fermions, following Fermi-Dirac statistics. In this case, quantisation is by anti-commutator relations.

Example. (Klein-Gordan fields must by quantised by commuator relations) The logical sequence is as follows:

(i) Micro-causality must follow, meaning that the measurement of two observables at spacelike separation cannot influence each other. For any observables $\hat{A}(x)$ and $\hat{B}(y)$, they must satisfy $[\hat{A}(x), \hat{B}(y)] = 0$ for $(x-y)^2 > 0$ and must be spacelike.

(ii) Observables are bilinear forms of field operators in quantum field theory. This means that commutation or anti-commutation of field operators at spacelike separation must vanish.

$$\begin{aligned}[XY, Z] =& X[Y, Z] + [X, Z]Y \\ =& X\{Y, Z\} - \{X, Z\}Y\end{aligned}$$

(iii) Quantised Klein-Gordan field was given as

$$\hat{\phi}(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}} (\hat{a}(\vec{p})e^{ip\cdot x} + \hat{a}(\vec{p})^\dagger e^{-ip\cdot x})$$

the first case is that the field, and consquently the mode operators commute, and the second case is that they anti-commute.

(iv) In the case that they commutator is used, in the equal time frame,

$$[\hat{\phi}(x), \hat{\phi}(y)] = \int \frac{d^3\vec{p}}{(2\pi)^3 2E_p} (e^{i\vec{p}\cdot(\vec{x}-\vec{y})} - e^{-i\vec{p}\cdot(\vec{x}-\vec{y})}) = 0$$

(v) In the case that the anti-commutator is used,

$$[\hat{\phi}(x), \hat{\phi}(y)] \neq 0 \qquad \& \qquad \{\hat{\phi}(x), \hat{\phi}(y)\} \neq 0$$